

MUHAMMAD ABDULLAH BUTT

Full Stack AI Engineer & AI Researcher

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PROFESSIONAL SUMMARY

Full Stack AI Engineer and AI Researcher with 3+ years of production experience building scalable web platforms and LLM-powered systems. Published author of MAPF-Lite, a published deepfake detection framework achieving 99.34% accuracy at 0.566M parameters, and a contributing author to a Springer Nature edited volume. Proven track record delivering end-to-end products from database schema to cloud deployment, with deep expertise in RAG pipelines, agentic AI, financial/market prediction models, and multimodal computer vision. Certified Oracle Generative AI Professional and IBM Agentic AI Expert.

PUBLICATIONS

[MAPF-Lite: A Parameter-Efficient Multimodal Framework for High-Fidelity Deepfake Detection](#) — FLINS-ISKE Journal, 2026 (Published)

Sole architect and lead researcher

- Achieved 99.34% accuracy and 99.84% AUC on FakeAVCeleb; 99.40% cross-dataset accuracy on DeepfakeTIMIT (zero-shot)
- Reduced trainable parameters by 99.6% to 0.566M using frozen CLIP/Whisper backbones with SRM and DCT analysis
- 4.7x real-time inference speed; 1.1 GB peak memory, deployable on edge and mobile devices
- Integrated explainable-AI (XAI) modules with visual heatmaps and frame-level suspicion scores for forensic transparency

[Machine Learning and Knowledge Engineering for Decision Making](#) — Springer Nature (Proceedings of the 17th International FLINS Conference & 21st ISKE Conference, FLINS-ISKE 2026, Sydney, Australia)

- Contributing author; research findings from the MAPF-Lite deepfake detection framework included in the conference proceedings volume, Part III

[“Deepfake Detection in the Multimodal Era: A Systematic Review of Techniques, Benchmarks, and Open Challenges”](#) — Review Paper (In Preparation)

- Lead author; systematic review surveying multimodal deepfake detection techniques, parameter-efficient architectures, and benchmark datasets, extending on findings from the MAPF-Lite framework

WORK EXPERIENCE

Senior Full-Stack AI Engineer

Mar 2026 - Present

Huzzle.com | London, UK (Remote)

- Developing AI-driven recruitment infrastructure integrating LLM pipelines with full-stack product features
- Building scalable backend services and frontend interfaces for a live, high-traffic B2C platform

Full Stack Engineer

Jan 2023 - Feb 2026

MAQ Enterprises Ltd. | Rawalpindi, Pakistan

- Engineered production web applications using Next.js, Node.js, and Prisma ORM; reduced average page load time by ~40% through SSR optimization and code splitting
- Designed RESTful APIs and optimized PostgreSQL and MongoDB schemas, improving data retrieval speed by ~35% on key endpoints
- Implemented CI/CD pipelines with Jenkins and Docker, cutting release cycle time by 30%
- Automated end-to-end testing with Cypress and Jest, reducing production bug incidence by 25%
- Mentored junior developers through structured code reviews, lifting team code-quality scores on internal audits
- Provisioned infrastructure-as-code using Terraform for consistent, repeatable environment deployments

AI Researcher

2024 - 2026

- Designed the MAPF-Lite multimodal architecture (CLIP + Whisper + ViT) for audiovisual deepfake detection, culminating in a peer-reviewed publication
- Implemented novel data augmentation and prompt-learning techniques to harden model robustness against adversarial multimedia attacks
- Developed lightweight attention mechanisms reducing computational overhead while preserving edge-device detection accuracy

Full-Stack Developer

2025 - May 2026

Estabraq (Fashion Brand) | Pakistan | Live site: estabraq.pk

- Architected a live fashion e-commerce platform using Next.js App Router, TypeScript, Prisma ORM, and Supabase achieving sub-2-second load times in production
- Integrated GraphQL APIs, Redis caching, and CDN-backed image optimization, reducing client-server latency by 30% and bounce rates measurably
- Automated deployment pipelines with Docker and CI/CD tooling, reducing release cycles by 25%
- Built companion React Native mobile app for omnichannel shopping experience (in progress)

KEY PROJECTS

ForexAI - Mobile App for Forex Learning & Market Prediction

React Native - XGBoost - Decision Trees - Kronos AI - Python

- Built a mobile application that teaches forex trading fundamentals alongside an AI-powered market prediction module
- Fine-tuned XGBoost and Decision Tree models on historical forex price data for short-term market movement prediction
- Integrated Kronos AI, a financial time-series foundation model, and ensembled its outputs with the fine-tuned XGBoost and Decision Tree models to improve prediction accuracy and robustness
- Designed an in-app learning curriculum pairing educational content with live prediction insights for retail traders

MAPF-Lite - Parameter-Efficient Multimodal Deepfake Detection Framework

PyTorch - Vision Transformers - CLIP - Whisper - SRM - DCT

- Designed and engineered a parameter-efficient multimodal architecture for high-fidelity audiovisual deepfake detection, achieving 99.34% accuracy and 99.84% AUC on FakeAVCeleb
- Reduced trainable parameters by 99.6% to 0.566M using frozen CLIP/Whisper backbones combined with SRM and DCT forensic analysis
- Achieved 4.7x real-time inference speed at 1.1 GB peak memory, enabling deployment on edge and mobile devices
- Built explainable-AI (XAI) modules producing visual heatmaps and frame-level suspicion scores for forensic transparency

TeleMed AI - Telemedicine Ecosystem

React Native - Firebase - Computer Vision

- Built a cross-platform telemedicine system with an AI-powered diagnostic scan for initial patient assessments
- Engineered multi-role architecture (patient, doctor, admin) with real-time slot booking and practitioner onboarding

LingumedAI - Clinical Decision Support for Nurses

React - FastAPI - LangChain - RAG - pgvector - OpenAI API

- Built an AI-powered clinical assistant giving nurses instant access to doctor-level diagnostic guidance, drug interaction checks, and protocol recommendations, reducing critical dependency on unavailable physicians
- Architected a domain-specific RAG pipeline ingesting medical literature, clinical guidelines, and drug databases into a pgvector store, enabling grounded, hallucination-resistant responses over sensitive healthcare data
- Engineered a React frontend optimized for high-stress clinical environments with low-latency streaming; FastAPI backend with real-time output via server-sent events
- Implemented role-based access control and audit logging to meet healthcare data governance requirements
- Designed explicit uncertainty escalation so the system flags ambiguous cases rather than generating confident but unsafe answers

TECHNICAL SKILLS

AI & LLM: LangChain, LlamaIndex, OpenAI API, Anthropic Claude, RAG Pipelines, Pinecone, pgvector, Agentic Workflows

Machine Learning & Fintech AI: XGBoost, Decision Trees, Kronos AI, Model Fine-Tuning, Ensemble Modeling, Time-Series Forecasting

Computer Vision: PyTorch, Vision Transformers, CLIP, Whisper, Model Quantization, PEFT, Edge Deployment

Frontend: Next.js, React, React Native, TypeScript, Tailwind CSS, GraphQL

Backend: FastAPI, Node.js, Prisma ORM, RESTful APIs, Redis, Django

Databases: PostgreSQL, MongoDB, Supabase

DevOps & Cloud: Docker, Jenkins, Terraform, CI/CD, Railway, Vercel, Oracle Cloud, AWS

EDUCATION

Bachelor of Science in Computer Science

Sep 2022 - Mar 2026

GIFT University, Gujranwala, Pakistan

- Migrated from University of Sialkot to GIFT University to continue and complete the degree
- Thesis: MAPF-Lite deepfake detection framework, published at FLINS-ISKE 2026 international journal

Bachelor of Science in Computer Science

Nov 2021 - Aug 2022

University of Sialkot, Sialkot, Pakistan

- Completed first year before migrating to GIFT University to continue the degree

CERTIFICATIONS

- OCI Certified Generative AI Professional - Oracle, 2025
- OCI Certified AI Foundations Associate - Oracle, 2025
- Agentic AI Expert - IBM
- Build RAG Applications - IBM, 2026
- NLP and Computer Vision Specialist - IBM
- Generative AI: Language Modeling with Transformers - IBM
- Advanced React - Meta
- Machine Learning Using Python - IBM